Assignment: Mini NLP Project

Alvin Mathew

24UBC211

NLP-Based Sentiment Analysis of Customer Reviews Using TF-IDF and Logistic Regression

Introduction

There is a lot of information in customer reviews about what people think and how they feel about products. Looking at a lot of reviews manually can take a long time. Natural Language Processing, which is often called NLP, can help with this. NLP can look at the text automatically. Find out if the feeling in the review is positive, negative or neutral. In this project a system for analyzing sentiment is created. This system sorts customer reviews into three groups: negative and neutral. The system uses text cleaning, a method called TF-IDF to get the important words and a model called Logistic Regression. The model is tested using Accuracy, Precision, Recall, F1-Score and a Confusion Matrix to see how well it works.

Problem Statement

Customer reviews give us insights, into what people think and feel about products.. Going through a lot of reviews by hand takes a lot of time.

The goal of this project is to create a system that uses natural language processing to automatically sort customer reviews into three types: negative or neutral.

The system works by cleaning the text turning words into values using TF-IDF and then using Logistic Regression to predict the sentiment of each review.

Objectives

The main objectives of this project are:

- To learn and prepare customer review text using natural language processing techniques.
- To turn reviews into numbers using TF-IDF.
- To build a sentiment classification model using Logistic Regression.
- To sort reviews into negative and neutral groups.
- To check the models performance using Accuracy, Precision, Recall, F1-Score and a Confusion Matrix.
- To look at the models results. Find out what cannot do well.

Dataset

The dataset used in this project is called Dataset-SA.csv. It holds customer reviews and the sentiment labels that go with them. At first the dataset had 205,052 records. When we prepared the data we took away missing review entries and duplicate records leaving 170,675 records.

The main attributes in the project are the customer review text and its sentiment label. The sentiment labels are positive, negative and neutral. Before turning the review text into numbers, with TF-IDF for classification we. Preprocessed the review text.

Data Preparation

We turned the review text into lowercase, removed characters and extra spaces cut out stopwords with NLTK and used the Porter Stemmer for stemming. We also removed duplicate records before doing analysis.

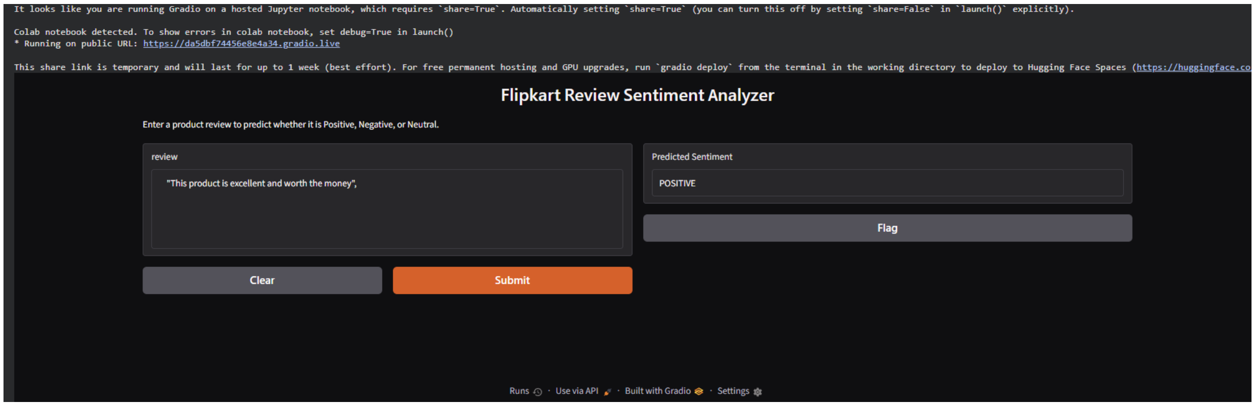
Methodology

Methodology, in this project has steps: text preprocessing → TF-IDF feature extraction → Logistic Regression classification → prediction → evaluation. First customer reviews were cleaned by converting the text to lowercase removing characters and unnecessary spaces removing stopwords and applying stemming. Next TF-IDF was used to turn the cleaned review text into features. These features were then fed into a Logistic Regression model to classify each review as positive, negative or neutral. Finally model predictions were checked with Accuracy, Precision, Recall, F1-Score and a Confusion Matrix. Methodology also shows the steps of cleaning, feature extraction, classification, prediction and evaluation.

Implementation

Implementation of the project was carried out using Python on Google Colab. I used Pandas, NumPy, NLTK, Scikit-learn, Matplotlib and Seaborn as the libraries. I applied NLP methods such as cleaning the text removing words reducing words to their roots and using TF-IDF to extract features. I chose Logistic Regression as the classification algorithm. I split the dataset into 80% training data and 20% testing data making sure the split was stratified so that the sentiment classes stayed balanced. I trained the model on the training data. Then tested it on the testing data. I measured its performance, with Accuracy, Precision, Recall, F1-Score and a Confusion Matrix.

A simple **Gradio-based user interface** was also created to demonstrate the practical use of the trained sentiment analysis model. That interface allows a user to enter a customer review and also receive the predicted sentiment as **Positive, Negative, or Neutral**.



Results

The Logistic Regression model achieved an **Accuracy of 89.52%**, **Precision of 84.88%**, **Recall of 89.52%**, and an **F1-Score of 86.82%** on the test data. The confusion matrix showed that the model performed well in identifying positive and negative reviews. However, the model did not correctly predict any neutral reviews, indicating that neutral sentiment was more difficult for the model to distinguish.

Metric	**Score**
Accuracy	89.52%
Precision	84.88%
Recall	89.52%
F1-Score	86.82%

The confusion matrix and sample predictions were used to further analyze the model's performance.

Limitations

The main limitation of the developed sentiment analysis model is its struggle to correctly identify reviews. The confusion matrix shows that the model failed to predict a neutral review correctly. Short or unclear reviews can also be hard to classify because they often lack sentiment cues. For example a review, like "It's okay nothing special" was misclassified as positive. The model depends on TF-IDF features, which do not capture the context or deeper meaning of sentences. As a result the model may not work well across all kinds of customer reviews.

Future Scope

The project can be improved in the future by using more diverse datasets that include a wider range of customer reviews. Better handling of ambiguous reviews could also improve classification performance. Additional preprocessing techniques and advanced NLP models could be explored to understand the context of reviews more effectively. The system could also be developed into an application where users can enter a review and receive its predicted sentiment automatically.

Conclusion

This project demonstrated the use of Natural Language Processing for sentiment analysis of customer reviews. The reviews were preprocessed using text cleaning, stopword removal and stemming and TF-IDF was used to convert the text into features. A Logistic Regression model was then trained to classify reviews, into negative and neutral categories. The model achieved an accuracy of **89.52%** with an F1-Score of **86.82%**. The results show that the approach can effectively classify customer reviews although identifying neutral sentiment remains a major limitation. Overall the project demonstrates how basic NLP techniques and machine learning can be applied to automatically analyze customer opinions.